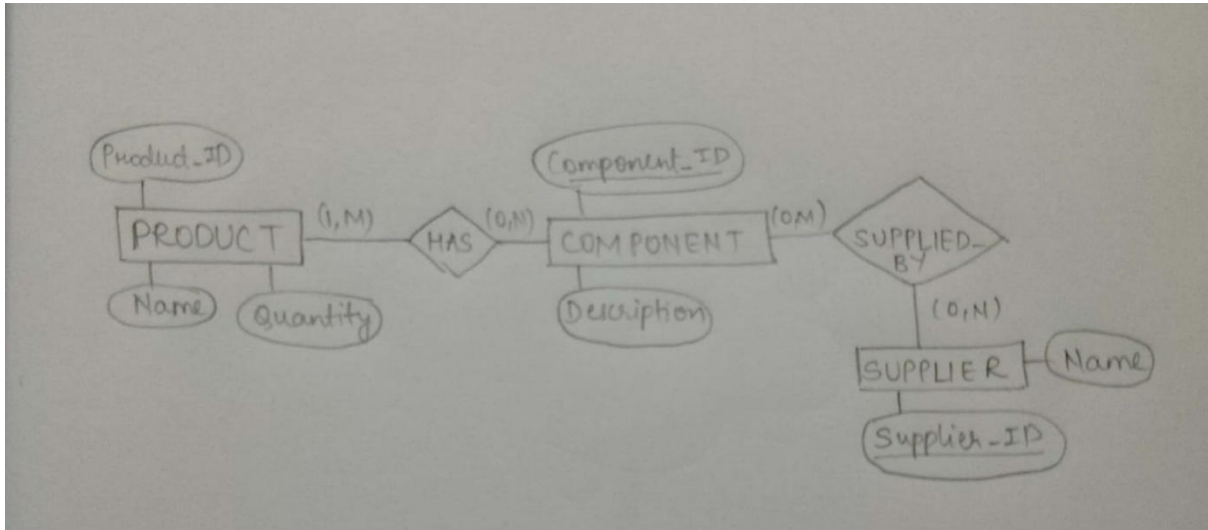


ASSIGNMENT – 3

Q.1 –

ER Model –



Relational Model –

Since , both the relationships are many to many we have to make a separate entity for these relationships as well . So , in total there will be 5 entities.

Entities:

1. PRODUCT(Product_ID , Name , Quantity)

PK : Product_ID

2. COMPONENT(Component_ID , Description)

PK : Component_ID

3. SUPPLIER(Supplier_ID , Name)

PK : Supplier_ID

4. PRODUCT_COMPONENT(Product_ID , Component_ID)

PK : (Product_ID , Component_ID)

FK : Product_ID references Product_ID(Product)

Component_ID references Component_ID (Component)

5. COMPONENT_SUPPLIER(Component_ID , Supplier_ID)

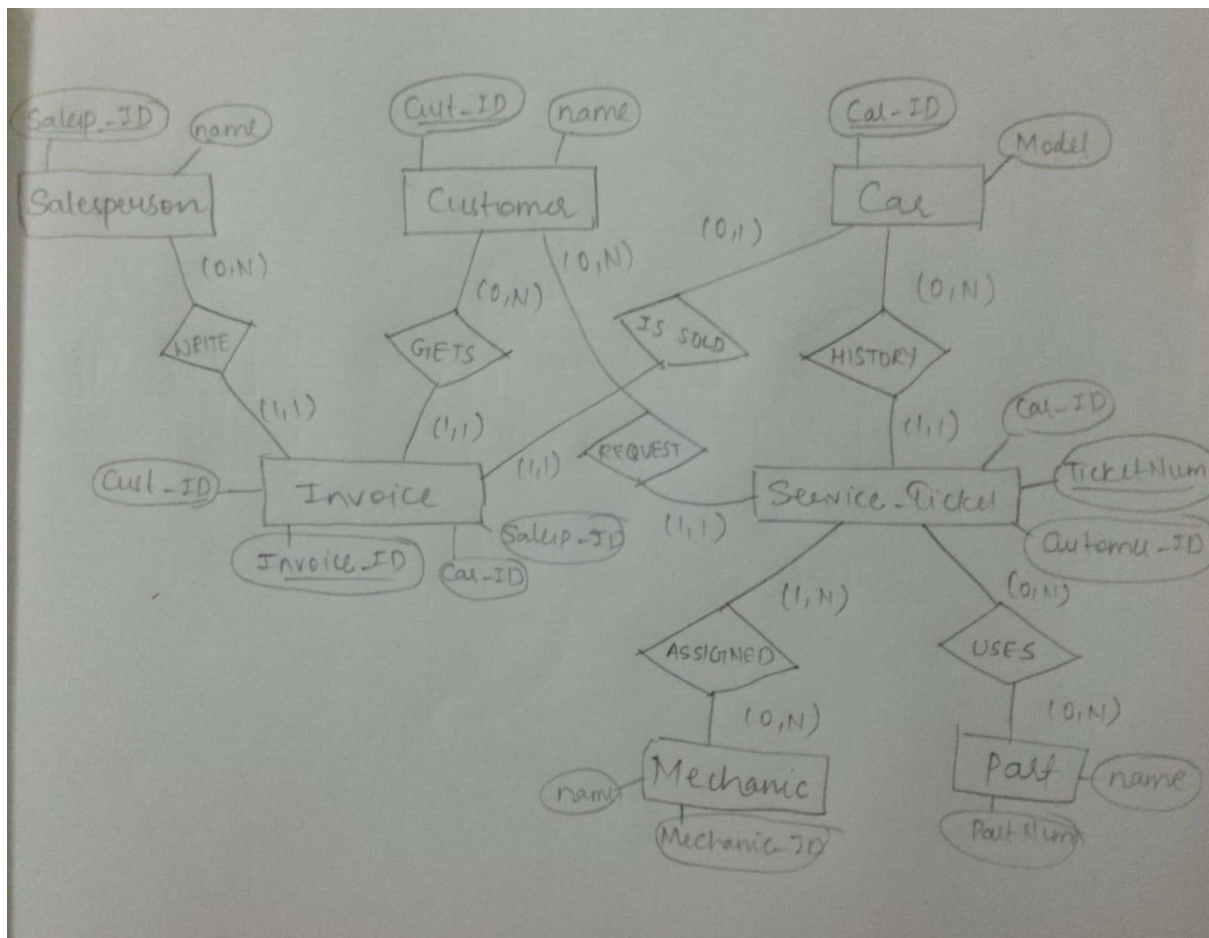
PK : (Component_ID , Supplier_ID)

FK : Component_ID references Component_ID (Component)

Supplier_ID references Supplier_ID(Supplier)

Q.2-

ER Model –



Relational Model –

There are 2 many to many relationships which will become the entities.

1. SALESPERSON(Salesp_ID , name)

PK : Salesp_ID

2. CUSTOMER(Cust_ID , name)

PK : Cust_ID

3. INVOICE(Invoice_ID , Cust_ID , Salesp_ID ,Car_ID)

PK : Invoice_ID

FK : Cust_ID references Cust_ID(Customer)

Salesp_ID references Salesp_ID (Salesperson)

Car_ID references Car_ID(Car)

4. CAR(Car_ID , Model)

PK : Car_ID

5. SERVICE_TICKET(TicketNum , Car_ID , Cust_ID)

PK : TicketNum

FK : Car_ID references Car_ID(Car)

Cust_ID references Cust_ID(Customer)

6. MECHANIC(Mechanic_ID , name)

PK : Mechanic_ID

7. PART(Part_ID , name)

PK : Part_ID

8. TICKET_MECHANIC(TicketNum , Mechanic_ID)

PK : (TicketNum , Mechanic_ID)

FK : TicketNum references TicketNum(SERVICE_TICKET)

Mechanic_ID references Mechanic_ID(Mechanic)

9. TICKET_PART(TicketNum , Part_ID)

PK : (TicketNum , Part_ID)

FK : TicketNum references TicketNum(SERVICE_TICKET)

Part_ID references Part_ID(Part)